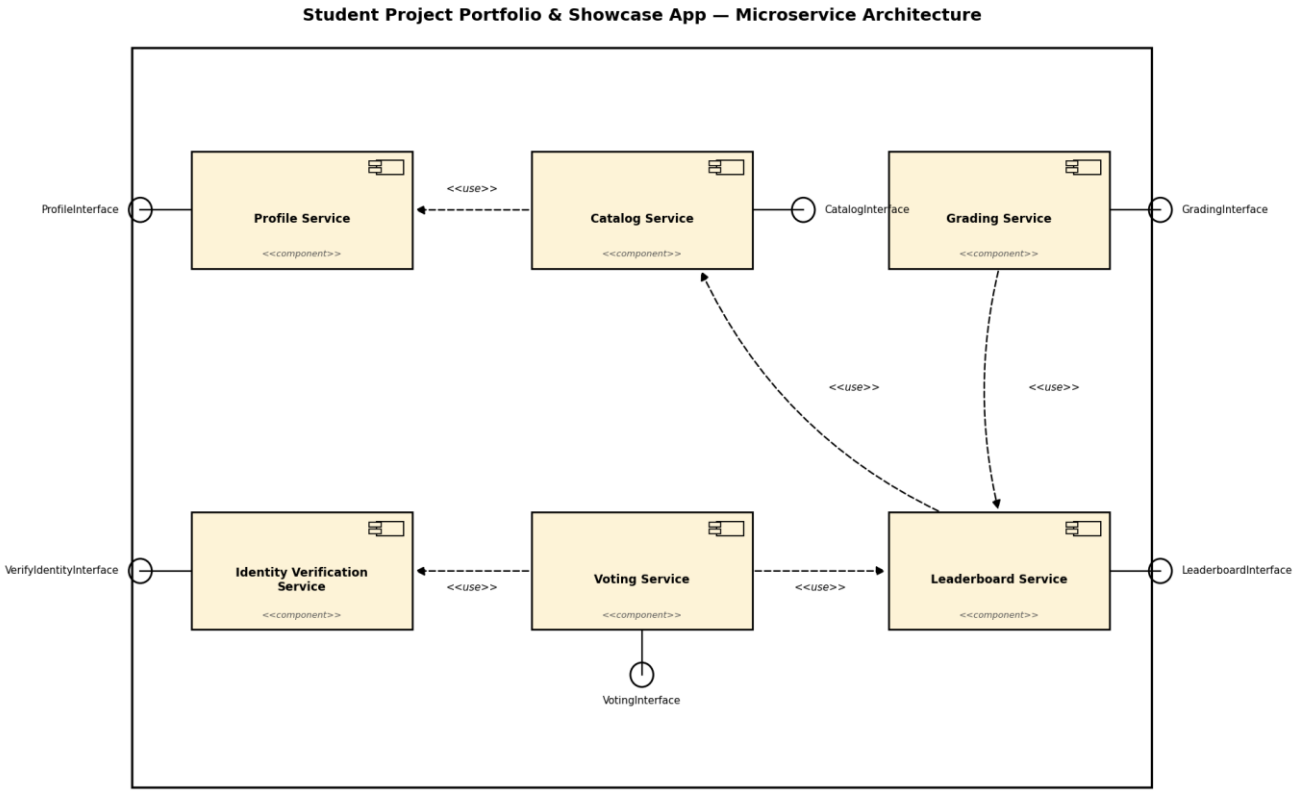
Lab 3: Component Diagram and Architecture Justification

Problem Statement #04: Student Project Portfolio & Showcase App

UML Component Diagram



Architecture Selection: We chose Microservices Architecture for the Student Project Portfolio & Showcase App.

Reason 1: Independent Scaling for Bursty, Uneven Load

The system's components face very different load patterns. The Voting Service and Leaderboard Service must handle a large, unpredictable burst of public traffic during the live expo window, while the Grading Service (a handful of judges) and Profile Service (student teams, submitted ahead of time) see steady, low volume. Microservices let each service scale independently to match its own demand, so voting traffic can be scaled out without over provisioning the rest of the system.

Reason 2: Independent Deployment and Fault Isolation

Profile creation, grading, and voting are built and refined on different timelines with different risk levels. As a set of independent services, each can be developed, tested, and deployed on its own schedule. Just as importantly, an issue in one service does not take down unrelated functionality; if the Identity Verification Service experiences a problem during the event, Catalog browsing and Profile viewing remain available.

Security Advantage

Isolating the Identity Verification Service as its own microservice lets it enforce a dedicated, stricter security boundary (OTP secret handling, rate limiting, audit logging) around the most attack sensitive part of the system, anti sybil vote verification, without exposing that attack surface to, or being constrained by, the rest of the application.

Performance Benefit

Because the Leaderboard Service is its own independently deployable and scalable component, it can be optimized and horizontally scaled specifically to meet the NFR-001 requirement of sub 300 ms real time updates, independent of slower, write heavy operations happening in the Grading or Profile services.